Antisolvent crystallization for nickel sulfate production: an 8000 t/y plant design and resource assessment

Abstract

An 8000 t/y nickel sulfate hexahydrate plant is conceptually designed for battery-material production using published crystallization and phase-equilibrium data and closed-loop Aspen Plus balances. The selected antisolvent configuration combines 75% feed-water removal, acetone crystallization, displacement washing and two-stage solvent recovery. At a prescribed 700 mg Ni/kg mother-liquor endpoint, it requires 241.8 kg fresh Ni and 60.3 kg makeup acetone per tonne of specified hexahydrate. Relative to MVR with downstream wash-liquor return, fresh-Ni demand falls by 13.6%, equivalent to 304 t/y. At the reference prices, including a 10 USD/kg Ni shadow value and zero non-hexahydrate Ni credit, the selected resource index falls from 3038.5 to 2893.7 USD/t, providing a 1.16 million USD/y margin for additional annualized equipment and operating costs. An acid–nickel balance explains the contribution of wash-return design. The antisolvent route retains its resource advantage up to approximately 1230 mg/kg residual Ni under conservative heat allocation, provided wash displacement is approximately 99.04% at the assigned dissolution. A separate feed-preheat target increases the annual resource margin to 1.50 million USD. The design combines improved nickel utilization with explicit solvent-recovery, washing and heat-transfer requirements, giving a defined process candidate for composition-specific product and equipment validation.

Keywords: nickel sulfate hexahydrate; antisolvent crystallization; acid balance; wash-liquor recycle; solvent recovery; mechanical vapor recompression

1 Introduction

Nickel sulfate hexahydrate is a precursor for nickel-containing cathode materials, and crystallization is a final separation step in nickel-refining flowsheets [1, 2]. Recovery and product quality depend on both precipitation and the subsequent solid–liquid separation. Experiments on acidic nickel liquors show that crystallization conditions affect cake moisture and impurity retention, while repulping can remove surface-associated material more readily than incorporated magnesium [3, 4, 5]. Nickel remaining in the mother liquor, nickel dissolved during washing and entrained acid consequently belong to the same process-design problem.

Antisolvent addition offers an alternative to evaporation. Solvent-displacement studies have examined dosing, seeding and crystal recycle with isopropanol [6]; Demirel et al. measured nickel-sulfate precipitation with acetone and isopropanol in pure-salt and post-ion-exchange liquors [7]. A plant solvent loop also circulates water, acid and dissolved nickel. Its crystallizer feed therefore differs from the initial laboratory solution, and its recovery requirement depends on what happens to residual nickel after filtration.

Plant-wide salt-recovery studies connect antisolvent selection to solvent recovery and compare evaporation, cooling and antisolvent routes on common production bases [13, 14]. Nickel sulfate adds acid-rich recycle, hydrate water and a washing constraint. Aqueous $NiSO_4$–H_2SO_4 phase data support water-only concentration and cooling [8, 9, 10], whereas mixed-solvent electrolyte prediction requires composition-matched interaction data [11, 12]. Mechanical vapor recompression (MVR), the competing aqueous route, exchanges external heat demand for compression work and heat-transfer area [17, 18, 15, 16].

This study develops an 8000 t/y nickel sulfate hexahydrate plant concept for battery-material production and evaluates acetone crystallization against MVR on the same production basis. The selected antisolvent design combines 75% feed-water removal, denoted P75, with solvent recovery and displacement washing. Acid and wash-return balances explain its resource performance, while

endpoint interpolation identifies the separation requirements needed to preserve its advantage. Temperature–enthalpy matching then defines the heat-transfer services behind the utility and economic assessment. Reference dose reductions and secondary sensitivity grids are retained in the Supplementary Information (SI).

2 Process model and assessment methods

2.1 Common production basis and configurations

The design capacity is 8000 t/y specified solid $NiSO_4 \cdot 6H_2O$ (HEX), produced at 1000 kg/h over 8000 h/y. Fresh-feed flow is adjusted to achieve that output. Battery-material use sets the intended application; the model resolves nickel, acid, solvent and water inventories, while trace-metal partition and final hydrate assays determine product qualification. Feed contains 4.544 g/L Ni and 0.100 mol/L formal H_2SO_4 at an assumed density of 1.020 kg/L. The nickel level follows the experimental post-ion-exchange liquor of Demirel et al. [7]; the modeled feed contains apparent $NiSO_4$, H_2SO_4, water and acetone, with minor metals omitted. The acid series changes formal acid at fixed nickel concentration and density, so acid replaces a small amount of feed water. Between 0.05 and 0.15 mol/L, this replacement is about 1% of the water inventory and contributes to the 1.1% decrease in the ASC recovery-column reboiler duty.

The comparison requires product formal acid below 50 ppmw; desolventization specifies 100 ppmw acetone and 0.05 wt% free water. These process screens accompany a fixed HEX output. Retained apparent non-HEX nickel receives zero credit in the main resource index; its inventory and valuation sensitivity are reported in SI Section S4. Impurity incorporation during nickel-sulfate crystallization is documented experimentally [5, 2].

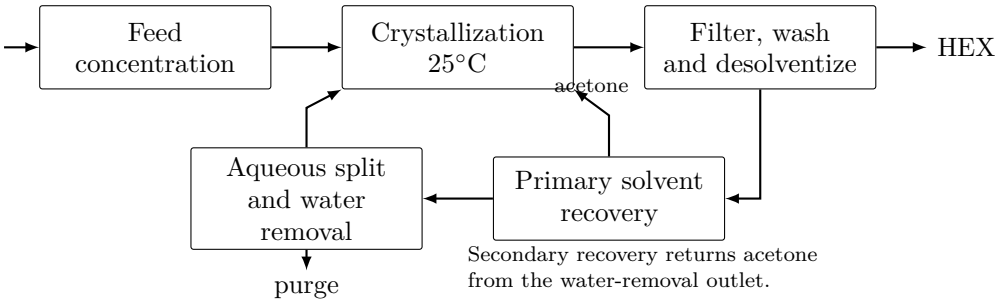
Table 1: Configurations used in the main comparison. Displacement and dissolution are simultaneous performance inputs; MVR has two equivalent wash contacts. ASC solvent is nominally 95 vol% acetone.

Specification	ASC W700	ASC P75	MVR
Fresh-feed water removal (%)	50	75	Aqueous endpoint
Mother-liquor Ni (mg/kg liquid)	700	700/1100/1400	31236
Wash/crystal ratio (kg/kg per contact)	0.60	0.60	0.40
Mother-liquor displacement (%)	99.00	99.05	98.00
Crystal dissolution (% per contact)	1.5	2.0	1/8.2/10
Crystal capture (%)	99.5	99.5	99.5
Retained liquid (kg/kg crystal)	0.20	0.20	0.20
Mother-liquor purge fraction (%)	50	50	50

The ASC process concentrates the fresh feed, cools it to 25°C and mixes it with recycled antisolvent. Filtration, displacement washing and desolventization produce HEX. A primary column recovers acetone before the aqueous recycle/purge split; a secondary column treats a water-removal outlet. Both preconcentration and recycle-water removal use vapor recompression. The primary recovery column has 20 equilibrium stages, feed stage 12 and 1.01325 bar top pressure; the secondary column has 20 stages, feed stage 10 and 0.6 bar top pressure. Their saved specifications and complete stream inventories are given in the SI.

MVR first concentrates the aqueous feed at 75°C and then reaches the 25°C endpoint of 0.934 mol $NiSO_4$ and 6.224 mol H_2SO_4 per kilogram free water, based on published aqueous phase data [10]. The two wash contacts use condensate. Upstream return combines mother and wash liquors before the 50% purge split; downstream return sends wash liquor directly to the recycle branch. Vapor from the 75 and 25°C services is compressed for condensation at 85 and 35°C. Isentropic, mechanical and motor efficiencies are 0.75, 0.95 and 0.95.

(a) ASC: concentration and solvent recovery



(b) MVR: wash return determines direct wash loss

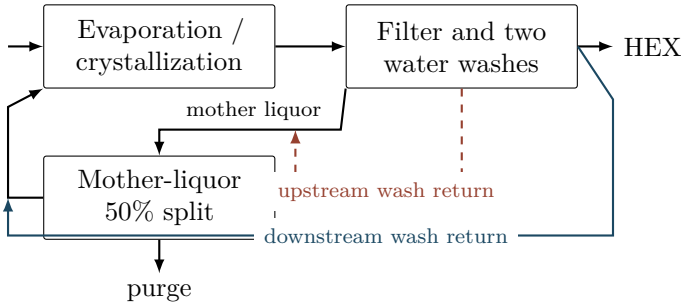


Figure 1: Process services and the wash-return alternatives. In (b), dashed and solid wash paths are alternative configurations. The mother-liquor split is fixed while the return location changes. Detailed ASC connections are in SI Section S2.

2.2 Property model and prescribed separation performance

The property framework is electrolyte NRTL–Redlich–Kwong with apparent-component balances mapped to Ni^{2+}, H^+, OH^-, HSO_4^- and SO_4^{2-} through the saved equilibrium chemistry. Solid HEX uses literature formation properties and heat capacity [19, 20]. Aqueous phase data constrain the water-only sections; the acetone–water binary set supplies the solvent-recovery VLE and mixing enthalpy. The model uses one liquid phase in crystallization and recovery. Composition-matched Ni–acid–acetone interactions and liquid–liquid stability remain experimental/model-development needs.

Mixed-solvent mother-liquor Ni is prescribed as a design variable. For pre-crystallization liquid mass M, nickel mass m_{Ni}, target liquid Ni mass fraction c and hydrate Ni mass fraction $w_{\mathrm{Ni},H}$, the crystal mass follows from conservation:

$$m_H = \frac{m_{\mathrm{Ni}} - cM}{w_{\mathrm{Ni},H} - c}. \tag{1}$$

An external controller sets crystallizer conversion, production scale and separation inputs, then resolves the recycle. The six-water hydrate stoichiometry is retained in formation and wash dissolution. Displacement and dissolution are specified independently so that their joint performance can be tested at the actual liquor composition. P75's 99.05% displacement is a near-threshold washer requirement, selected to pass the 50 ppmw acid screen with the assigned 2% dissolution.

The original acetone–water parameter set gives mean absolute errors of 0.433 K in bubble temperature, 0.00552 in vapor acetone mole fraction and 644.5 J/mol in mixing enthalpy against the BAC8 compilation [21]. An alternative set improves caloric agreement and is propagated through the reference dose comparison. Separate paired acid-dielectric calculations give no resolved resource-index change at the saved precision. Data, parameter fits and numerical acceptance checks are documented in SI Sections S1.1, S1.2 and S6.

2.3 Resource index and heat integration

Resource demand is compared using

$$J = p_H Q_H + p_E E + p_C Q_{CW} + p_A m_A + p_N m_N, \tag{2}$$

where heat and process cooling-water duties are in GJ/h, electricity in kWh/h, and acetone makeup and fresh Ni in kg/h. Reference values are 8 USD/GJ heat, 0.10 USD/kWh electricity, 0.20 USD/GJ cooling, 0.90 USD/kg acetone and a 10 USD/kg Ni shadow value. This selected operating/resource index supplies a margin for subsequent equipment, auxiliary and treatment costs. Product output and non-HEX credit remain fixed when prices change.

A heat cascade uses a 10 K minimum approach, 120°C steam and 20–30°C cooling water. Refrigeration evaporates at 15, 5 and −5°C and condenses at 40°C; electricity is 35% of Carnot performance. Connected exchangers are credited once. Temperature–enthalpy profiles separate compression superheat, latent heat and sensible cooling. Segmented areas use

$$A = \int_0^Q \frac{\mathrm{d}q}{U[T_h(q) - T_c(q)]}, \qquad U = 700\ \mathrm{W\,m^{-2}K^{-1}}. \tag{3}$$

For unprofiled ASC condensation intervals, favorable and unfavorable endpoint placements give two heat allocations. The conservative allocation is the nominal basis for P75 endpoint requirements. Composition-specific W700 and P75 profiles at 700 mg/kg refine their absolute indices. A separate preheat target holds column product states fixed, moves an equal duty from reboiling to feed heating, and cools purge to a common 25°C outlet. The cascade then reallocates available heat at its corresponding temperature.

2.4 Performance requirements and verification

The requirement c_{req} is the interpolated mother-liquor endpoint at which P75 and downstream-return MVR have equal J, subject to the product-acid screen. Interpolation uses complete P75 flowsheets at 700, 1100 and 1400 mg/kg with identical washing inputs. Price dependence follows directly from their saved fresh-Ni requirements:

$$\Delta J(c, p_N) = \Delta J(c, 10) + (p_N - 10)[m_{N,\mathrm{P75}}(c) - m_{N,\mathrm{MVR}}]. \tag{4}$$

Crossings are reported within the solved endpoint range. The preheat sensitivity subtracts the 700 mg/kg incremental saving of 41.51 USD/h from each P75 endpoint difference; it tests a constant-saving assumption separately from the complete-flow interpolation.

Accepted flowsheets pass recycle convergence, production, component/element and unit-energy checks, followed by save, reopen, clear and rerun verification. Inputs, numerical tolerances and save/reopen checks are retained with each accepted case in the SI and data archive.

3 Results and discussion

3.1 Selected plant design and resource demand

The P75 plant produces 8000 t/y specified HEX from approximately 53.2 $\mathrm{m^3/h}$ of the modeled purified nickel liquor. Its selected design integrates 75% fresh-feed water removal, acetone crystallization at 25°C, filtration and displacement washing, solvent recovery and aqueous recycle. At 700 mg/kg mother-liquor Ni, the fresh requirements are 241.76 kg Ni and 60.32 kg acetone per tonne of HEX (Table 2). Recovery of fresh Ni into the specified HEX is 92.36%, compared with 79.82% for downstream-return MVR.

The principal advantage is lower nickel demand: 38.00 kg/t, or 304 t/y at design capacity. P75 also requires less selected electricity, 2.025 versus 2.400 MW, while using 7.439 MW external

heat and circulating acetone. The resource index combines these competing demands. At the reference prices it is 2893.7 USD/t for P75 and 3038.5 USD/t for MVR, a 4.8% reduction. The following sections explain the recycle and solvent-design decisions that produce this margin and the separation performance required to retain it.

Table 2: Selected plant design and the improved MVR comparator at 8000 h/y. P75 uses the prescribed 700 mg/kg endpoint and joint washing specification. Both indices use the reference prices, zero non-HEX Ni credit and composition-specific heat targets; preheat is a separate design target. Electricity includes modeled compression, pumps and refrigeration.

Quantity	P75 antisolvent	Downstream-return MVR
Specified HEX production (t/y)	8,000	8,000
Fresh-feed volume (m^3/h)	53.20	61.57
Fresh Ni (kg/t HEX)	241.76	279.76
Ni recovery into specified HEX (%)	92.36	79.82
Acetone makeup (kg/t HEX)	60.32	0.00
External heat target (MW)	7.439	0.000
Selected electricity (MW)	2.025	2.400
Selected resource index (USD/t HEX)	2893.7	3038.5
Annual resource index (million USD/y)	23.15	24.31

3.2 Acid disposal couples wash losses to fresh-nickel demand

The reference W700 inventory identifies different roles for water, acid and acetone recycle. Its purge carries 121.04 kg/h of the 127.22 kg/h fresh acetone requirement, while the final secondary-recovery outlet carries 5.86 kg/h. The primary reboiler supplies 14.56 MW, or 94% of gross block heating. Acid disposal controls how much residual nickel leaves the circulating liquor; solvent recovery controls how much acetone reaches that outlet. The complete stream and equipment balances are in SI Sections S3 and S4.

For the aqueous route, define r as the mother-liquor mass ratio of Ni to formal H_2SO_4, and k as the fresh-feed acid/Ni mass ratio. Let N and A be Ni and formal-acid mass flows in product plus fines. The external balances give

$$\dot{m}_{\mathrm{Ni,feed}} = \frac{N + W - rA}{1 - rk}. \tag{5}$$

For wash liquor returned before the 50% purge split, $W = 0.5(\dot{m}_{\mathrm{Ni,wash}} - r\dot{m}_{\mathrm{acid,wash}})$; downstream return makes $W = 0$. At the reference feed, $1 - rk = 0.8062$. Holding N, A, r, k fixed gives $\Delta\dot{m}_{\mathrm{Ni,feed}} = 1.240\Delta W$: replacing lost nickel also introduces acid, which must leave through the same constrained outlet. Independent material closures reproduce this relationship over 76 acid, washing and routing scenarios.

Increasing feed acid from 0.05 to 0.15 mol/L raises upstream-return MVR fresh Ni from 251.65 to 320.42 kg/h and J from 2731.9 to 3478.1 USD/h. W700 remains near 257 kg/h Ni because its prescribed mixed-solvent endpoint gives a much lower Ni/acid disposal ratio. Its conservative index changes from 3349.7 to 3343.8 USD/h; the resource ordering reverses near 0.13 mol/L feed acid. All three solved feed compositions pass the product-acid screen. This crossing concerns upstream-return MVR; the P75 comparison below uses downstream return.

Moving the wash return after the purge lowers MVR fresh Ni by 2.13 kg/h at 1% dissolution per contact and by 21.14 kg/h at 8.2%. At the latter point, J falls by 226.0 USD/h to 3128.6 USD/h while product acid remains 32.35 ppmw (Fig. 2). The 10% downstream-return case needs 289.78 kg/h Ni and gives $J = 3147.3$ USD/h with 33.55 ppmw acid. Thus return location, rather than dissolution alone, determines the direct wash-loss penalty. The remaining increase with dissolution includes the changing retained product salt and circulating loads.

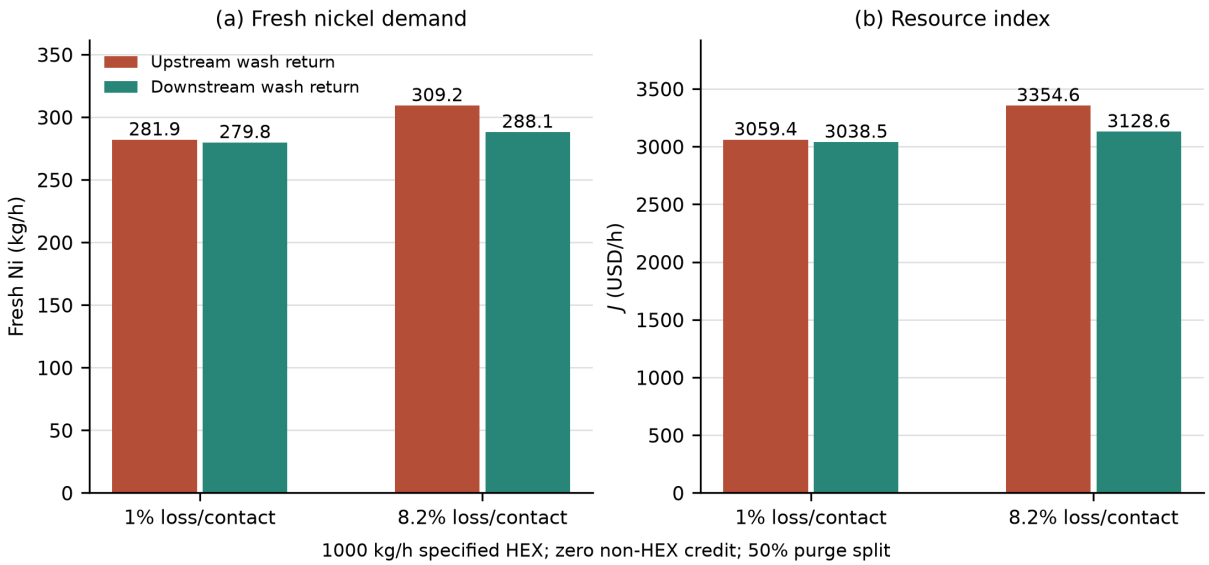


Figure 2: Complete-flow effect of the MVR wash-liquor return location at the reference 0.1 mol/L feed. The mother-liquor split remains 50%, with 98% wash displacement per contact and 1000 kg/h specified HEX. At each dissolution level, both return arrangements have the same product composition. The resource index assigns zero non-HEX credit.

3.3 Solvent recovery and preconcentration

Strengthening primary acetone recovery acts before the aqueous purge. Raising recovery from 99.4956 to 99.95% lowers acetone makeup from 127.22 to 12.90 kg/h. The unrecovered fraction falls from 0.5044 to 0.05%, explaining the 89.9% makeup reduction. Fresh Ni changes by only about 0.02 kg/h. The primary reboiler rises from 14.558 to 14.650 MW while secondary-recovery duty falls; the full-flow resource saving is 100.6–102.0 USD/h under matched endpoint heat allocation. This recovery case is a separation-performance target for the specified column.

P75 reduces the water and solvent circulated through crystallization and recovery. At a 700 mg/kg endpoint, fresh Ni and acetone fall to 241.76 and 60.32 kg/h. The resulting mother liquor contains 55.79 wt% acetone on an acetone-plus-free-water basis and 0.6048 mol formal acid/kg free water. These are the composition coordinates for testing P75; acetone on the total-liquid basis is 54.27 wt%.

The more concentrated acid inventory makes washing decisive. With 99% displacement and 1.5% dissolution, the 75% preconcentration case gives 51.61 ppmw product acid. P75 instead specifies 99.05% displacement and 2% dissolution at 0.60 kg wash/kg crystal, giving 49.23 ppmw. Scaling residual original-liquor acid locally places the minimum displacement near 99.035%. The 0.77 ppmw margin identifies a near-threshold washer requirement. Acetone-rich wash liquor tends to suppress salt dissolution; the assigned 2% loss is a stress assumption whose magnitude should be measured together with displacement.

Actual-composition condenser profiles give $J = 3346.7$ USD/h for W700 and 2893.6 USD/h for P75 at 700 mg/kg. Their residual heat-placement spans are below 0.1 USD/h. P75 therefore has a 144.8 USD/h margin over downstream-return MVR at $J = 3038.5$ USD/h; W700 remains 308.3 USD/h higher. SI Table S34 distinguishes the endpoint, profiled-condensation and preheat calculation levels.

The original 10% acetone-dose reduction gives a much smaller resource change: W700 saves approximately 67 USD/h relative to BASE95 at equal 700 mg/kg endpoints. Raising residual Ni removes that saving near 803 mg/kg with the original binary parameters and 795 mg/kg with the alternative set. Those comparisons share the reference recovery specification and a different comparator from P75; their detailed balances are retained in SI Section S7.

3.4 Economic margin and required separation performance

The complete P75 points at 700, 1100 and 1400 mg/kg require 241.76, 250.82 and 258.09 kg/h fresh Ni. Their conservative ΔJ values relative to downstream-return MVR are −144.7, −36.5 and +50.4 USD/h. Linear interpolation gives 1226 mg/kg, reported as about 1230 mg/kg. The favorable endpoint allocation gives 1296 mg/kg. The 700 mg/kg composition-specific calculation lies within 0.2 USD/h of the conservative result, supporting that allocation as the nominal estimate; the 1100 and 1400 mg/kg points retain endpoint heat placement. The SI preserves both allocations.

Nickel valuation shifts this requirement more than the heat-allocation range. At $p_N = 6.2$, 8, 10 and 12 USD/kg, conservative parity occurs at approximately 700, 1005, 1226 and 1379 mg/kg, respectively (Fig. 3). At 13 USD/kg, P75 remains below MVR across the evaluated 700–1400 mg/kg range. These changes follow from the lower fresh-Ni demand of P75; all other prices and saved states are fixed. Product acid remains 49.11–49.23 ppmw across the three solved endpoints, so displacement performance remains the principal acid constraint.

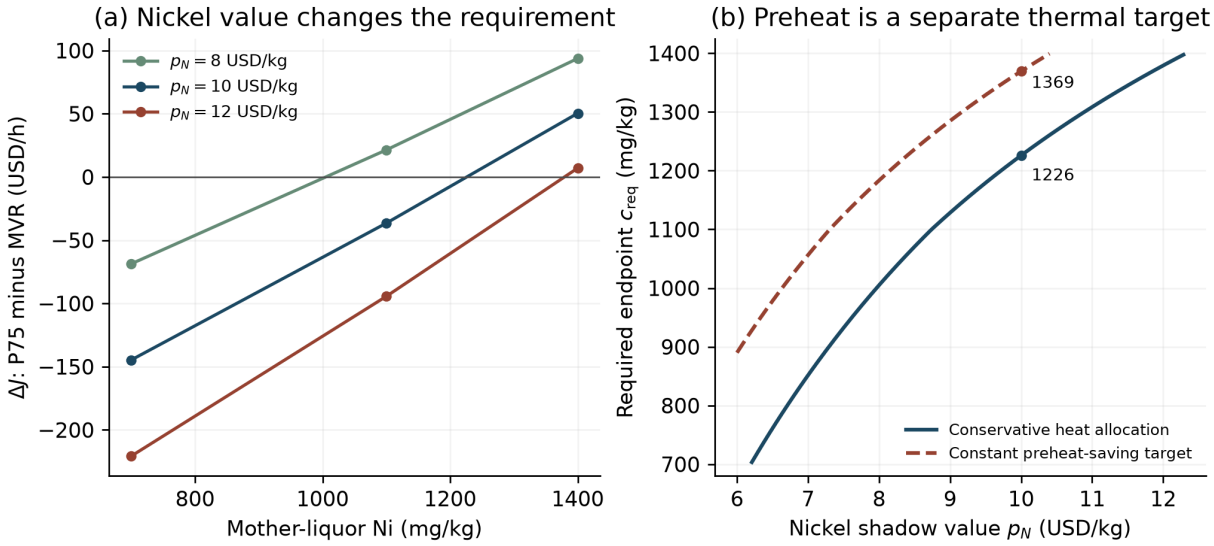


Figure 3: P75 performance requirements relative to downstream-return MVR at 1% dissolution, 1000 kg/h HEX and zero non-HEX Ni credit. (a) Saved complete-flow states with conservative endpoint heat allocation at three Ni shadow values. (b) Piecewise-linear crossings within 700–1400 mg/kg; the dashed curve subtracts a constant 41.51 USD/h preheat-saving target from each endpoint. Other prices are fixed as in Eq. 2. P75 uses 75% feed-water removal, 99.05% wash displacement and 2% dissolution. Curves stop at the evaluated endpoint limits.

The P75 RECOV feed can receive 0.984 MW from condensation and 0.422 MW from purge cooling, reaching 57.42°C with a 10 K minimum approach. On a common 25°C purge-outlet basis, this 1.406 MW preheat target saves 41.51 USD/h. Carrying that incremental saving unchanged across the three endpoint differences moves parity from 1226 to 1369 mg/kg at $p_N = 10$ USD/kg. This is a thermal-design sensitivity; the associated column product states are fixed. P75's complete-flow comparison retains its approximately 99.5% primary recovery, so the separate 99.95% recovery result is not added to the nominal endpoint requirement.

Published crystallization data place these requirements in a useful experimental range. Converting the pure-salt acetone series to solvent-basis acetone and total-liquid Ni gives approximately (30%, 13.6 g/kg), (47%, 3.8 g/kg) and (64%, 0.5 g/kg). Under the stated closed-batch and initial-water assumptions, logarithmic interpolation at P75's 55.79 wt% acetone gives about 1.33 g/kg. A separate illustrative slope range anchored to the post-IX B endpoint gives about 1.00–1.25 g/kg. SI Table S26 reports the conversions, B/E points and assumptions. The 1.23 g/kg

nominal requirement and 1.37 g/kg preheat sensitivity are both close to this literature scale. The 0.70 g/kg reference is a more demanding endpoint. The decisive crystallization test is therefore at P75's solvent composition and acid molality, with the required wash performance measured on the resulting cake.

3.5 Heat integration and design implementation

The MVR heat target is supported by explicit temperature–enthalpy matches. At the reference upstream-return state, 40.424 MW is matched with at least 10 K local approach and about 5458 m^2 additional area at the stated U. Downstream return at 1 and 10% dissolution requires approximately 5446 and 5654 m^2, respectively, with zero external heat and selected electricity of 2.400 and 2.486 MW. These areas cover added vapor-to-process matches; existing feed recovery and final coolers are outside that area total. SI Sections S10 and S5 give the service-by-service profiles and pressure assumptions.

Temperature lift is consequential. At fixed MVR pressures, increasing the required approach from 10 to 15 K raises the external-heat target from zero to 32.57 MW, identifying a compressor-pressure redesign requirement. Finisher suction is about 595000 m^3/h near 20 mbar, so compressor maps, pressure losses and non-condensable removal are important equipment inputs. The downstream-return feed exchanger has a near-saturated hot inlet; the independently calculated water enthalpy drop agrees with Aspen within 0.12%, while local pressure and subcooling remain equipment specifications.

W700's four connected exchangers recover 28.039 MW and require about 2542 m^2 from segmented profiles. Two further feed-preheat matches total 3.008 MW and 213 m^2; P75's corresponding matches require 1.406 MW and 99 m^2. At fixed column product states, the incremental resource savings are 88.80 and 41.51 USD/h. The common-outlet and preheat targets are tabulated separately in SI Table S34.

At 8000 h/y, P75's profiled 700 mg/kg resource margin allows about 1.16 million USD/y of additional annualized ASC-specific cost relative to downstream-return MVR. Including the preheat target raises that allowance to about 1.50 million USD/y. Equipment assessment must charge the solvent-recovery columns, acetone inventory and containment, flammable-solvent service and larger preconcentration duty against this margin, with differential MVR equipment costs treated on the same basis.

Product and equipment development. The remaining design tests are concentrated at the selected P75 composition. The normalized pure-salt series has approximately 0.43 mol acid/kg water, compared with 0.60 in P75, and post-IX acid is unreported. Crystallization measurements should therefore establish residual Ni and liquid-phase stability at the P75 composition; washing tests should measure displacement and dissolution together. Trace-metal removal and hydrate preservation during solvent removal connect this process concept to the intended battery-material product. Exchanger coefficients, compressor maps and solvent-service equipment costs determine how much of the calculated annual resource margin can be retained in the installed plant.

4 Conclusions

An 8000 t/y nickel sulfate hexahydrate plant concept was developed for battery-material production by combining published crystallization and aqueous phase data with closed-loop process balances. The selected P75 antisolvent design integrates 75% feed-water removal, acetone recovery and displacement washing. At its prescribed 700 mg/kg endpoint, it reduces fresh-Ni demand by 13.6%, or 304 t/y, relative to MVR with downstream wash return.

At the reference resource prices, the selected index decreases by approximately 145 USD/t HEX, providing 1.16 million USD/y for additional annualized ASC costs. The feed-preheat target raises that margin to 1.50 million USD/y. Retaining the nominal advantage requires residual Ni below about 1230 mg/kg at a 10 USD/kg Ni shadow value, together with approximately 99.04% wash displacement at the assigned dissolution. These conditions define a specific antisolvent plant candidate with improved nickel utilization and a quantified economic margin; composition-matched product tests and equipment costing are the next design gates.

Data availability

The accompanying archive contains accepted Aspen cases, unsuccessful-run records, full resource ledgers and reproduction scripts. The entry point `reproduce_study.py` rebuilds the saved-state analyses, including the plant resource summary, price-dependent P75 requirements and literature normalization. The SI contains detailed model specifications, property checks, secondary sensitivities and numerical verification.

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